

MI-Driven MoE LoRA for Multi-Modal Maritime Segmentation

Anonymous CVPR submission

Paper ID ****

Abstract

001 *Adapting vision foundation models to multi-modal seman-*
002 *tic segmentation is challenging because these models are*
003 *designed for single-modal input, and naively fusing hetero-*
004 *geneous modalities often fails when sensor quality varies*
005 *across conditions. We propose a method that adapts SAM2*
006 *via MemorySAM’s memory-based cross-modal aggregation*
007 *and addresses two key failure modes of multi-modal fusion*
008 *with foundation models. First, when Mixture-of-Experts*
009 *(MoE) LoRA layers are used for modality-adaptive encod-*
010 *ing, soft gating causes all experts to converge to a uni-*
011 *form routing distribution, producing modality-agnostic fea-*
012 *tures that prevent downstream fusion from distinguishing*
013 *sensor quality. We resolve this by maximizing the mutu-*
014 *al information (MI) between modality identity and ex-*
015 *pert routing, driving each modality to utilize distinct ex-*
016 *perts and producing differentiated backbone features. Sec-*
017 *ond, the segmentation objective alone provides insuffi-*
018 *cient gradient signal for the fusion head to learn input-*
019 *adaptive gating weights. We introduce quality-aware cross-*
020 *modal scoring—enriching each modality’s representation*
021 *with multi-pool statistics through modality-specific com-*
022 *pression networks—combined with gating oracle supervi-*
023 *sion via KL divergence from per-modality auxiliary seg-*
024 *mentation heads. The resulting weights are applied with*
025 *temperature-scaled normalization for memory modulation*
026 *and as raw softmax for output fusion. To bridge the day-*
027 *night domain gap, we apply aggressive RGB corruption*
028 *augmentations including extreme darkening, complemen-*
029 *tary random masking [11], and stochastic full-channel ze-*
030 *roing. We validate each design choice on the MULTIAQUA*
031 *benchmark [6], achieving 94.00% mIoU on the daytime val-*
032 *idation set.*

033 1. Introduction

034 Reliable scene understanding is a critical capability for
035 unmanned surface vehicles (USVs) operating in mar-
036 itime environments. Unlike structured road scenes, wa-
037 terways present unique challenges: water surfaces ex-

hibit highly variable reflectance, obstacles appear at un-
predictable scales and distances, and visibility conditions
can shift dramatically between day and night. Multi-
modal sensing—combining RGB cameras with thermal and
LiDAR sensors—offers a principled approach to address
these challenges, as each modality provides complemen-
tary information robust to different environmental degrada-
tions [14, 15].

However, effectively fusing heterogeneous modalities
remains an open problem. Traditional multi-modal seg-
mentation methods [5, 9, 15] design modality-specific en-
coders or fusion modules, which scale poorly as the num-
ber of modalities increases. Meanwhile, foundation models
such as SAM [4] and SAM2 [8] have demonstrated remark-
able segmentation capabilities through large-scale pretrain-
ing, but they are fundamentally designed for single-modal
(RGB) input and instance-level tasks. Bridging foundation
models to multi-modal semantic segmentation thus requires
addressing two fundamental gaps: enabling multi-modal
input processing and transitioning from instance-level to
semantic-level understanding.

Recently, MemorySAM [1] proposed an elegant solu-
tion by reinterpreting SAM2’s temporal memory mech-
anism for cross-modal aggregation. By treating each
modality as a sequential “frame”, the memory bank nat-
urally captures modality-agnostic features, while a Se-
mantic Prototype Memory Module (SPMM) guides the
model toward semantic-level segmentation. Concurrently,
MoE-LoRA [16] introduced Mixture-of-Experts routing
into SAM’s Low-Rank Adaptation [3] layers, enabling
modality-adaptive parameter specialization with minimal
overhead.

In this work, we build upon these advances and investi-
gate how to effectively adapt foundation models for multi-
modal maritime segmentation under varying sensor condi-
tions. We evaluate on the MULTIAQUA dataset [6], which
provides synchronized RGB, thermal, and LiDAR data cap-
tured from USVs across daytime and nighttime conditions,
with four semantic classes: static obstacle, dynamic ob-
stacle, water, and sky. A key difficulty is that models are
trained on daytime data but must generalize to nighttime

conditions where RGB sensors become severely degraded.

Our approach adapts the MemorySAM framework to the maritime domain with the following design choices:

(1) Soft-MoE LoRA with MI-driven expert specialization. MoE-LoRA [16] employs hard top- k routing, where only k out of N experts are activated per token. When the number of experts is small (e.g., 3 experts matching 3 modalities), top- k routing discards significant expert capacity. Following insights from Soft Mixtures of Experts [7], we adopt soft gating where all experts contribute via softmax-weighted combination. However, we observe that soft gating causes expert routing to converge to a near-uniform distribution, producing modality-agnostic backbone features. To address this, we introduce an MI routing loss that maximizes the mutual information between modality identity and expert selection, driving each modality to route through distinct experts.

(2) Quality-aware cross-modal scoring with gating oracle supervision. Prior approaches [1] estimate each modality’s quality independently, which can suffer from sigmoid saturation—when all logits are large, scores converge to ~ 1.0 regardless of actual quality differences. We replace independent scoring with a cross-modal fusion head that jointly compares all modality features, producing softmax-normalized relative weights. To enrich the feature representation, we extract quality-sensitive statistics (global average pooling, global max pooling, and channel-wise standard deviation) through modality-specific compression networks. Crucially, the segmentation objective alone provides insufficient gradient signal to drive adaptive gating. We therefore introduce per-modality auxiliary segmentation heads that produce independent per-image quality estimates; oracle gating weights are derived from these estimates and used to supervise the fusion head via KL divergence. The learned weights are applied through two paths: temperature-scaled for memory modulation (UAMM) and as raw softmax for output fusion (AMF).

(3) Night-robust RGB augmentation. To bridge the day-night domain gap using only daytime training data, we apply aggressive RGB corruption including extreme brightness reduction (factor 0.03–0.25), complementary random masking [11] that zeros out 20–50% of spatial regions, and stochastic full-channel zeroing simulating complete RGB failure. These augmentations encourage the model to rely on thermal and LiDAR when RGB is degraded.

We empirically validate each component through controlled experiments and report results on the MULTIAQUA benchmark. The remainder of this paper is organized as follows: Sec. 2 reviews related work, Sec. 3 details our method, and Sec. 4 presents experimental results.

2. Related Work

2.1. Multi-Modal Semantic Segmentation

Multi-modal semantic segmentation aims to leverage complementary information from heterogeneous sensors to improve scene understanding beyond what any single modality can achieve. Early approaches [2, 12] adopted dual-encoder architectures with element-wise or concatenation-based fusion, but these methods are typically designed for fixed modal pairs (e.g., RGB-Depth or RGB-Thermal) and cannot easily generalize to arbitrary combinations.

Transformer-based architectures have significantly advanced this field. SegFormer [13] introduced a hierarchical Transformer encoder with a lightweight MLP decoder, eliminating positional encoding to enable flexible multi-scale feature extraction. Building on this backbone, CMX [14] proposed a unified cross-modal fusion framework with a Cross-Modal Feature Rectification Module (CM-FRM) and a Feature Fusion Module (FFM) that performs long-range context exchange across modalities, achieving state-of-the-art results across RGB-Depth, RGB-Thermal, RGB-Polarization, and RGB-LiDAR benchmarks. CMNeXt [15] further scaled this paradigm with the Self-Query Hub (SQ-Hub) and Parallel Pooling Mixer (PPX), enabling fusion across an arbitrary number of modalities with negligible per-modality parameter overhead and demonstrating +9.10% mIoU improvement with quad-modal input on the DELIVER benchmark.

StitchFusion [5] took a different approach by integrating large-scale pretrained models as both encoders and feature fusers, using a MultiAdapter module for cross-modal information transfer during encoding. This enables multi-modal fusion with minimal additional parameters by leveraging pretrained representations directly. MMSFormer [9] introduced a novel fusion block within a Transformer architecture that progressively improves performance as additional modalities are incorporated, demonstrating the effectiveness of modality-agnostic fusion on material and semantic segmentation benchmarks.

For modality robustness, Shin *et al.* [11] proposed Complementary Random Masking (CRM) for RGB-Thermal segmentation, which strategically masks input modalities during training to prevent over-reliance on any single sensor. Combined with a self-distillation loss, CRM forces the network to learn complementary representations, improving robustness when individual modalities are degraded. Our night augmentation pipeline draws inspiration from this approach, extending it with extreme brightness corruption and full-channel zeroing tailored for maritime nighttime scenarios.

For maritime environments specifically, the MULTIAQUA dataset [6] provides synchronized RGB, thermal, and LiDAR data across daytime and nighttime conditions.

It introduces robust training strategies that enable networks trained solely on daytime images to generalize to nighttime scenarios, highlighting the importance of augmentation-based domain adaptation for multi-modal maritime perception.

2.2. Foundation Model Adaptation for Segmentation

The Segment Anything Model (SAM) [4] established a new paradigm for promptable image segmentation through large-scale pretraining on over 1 billion masks. Its successor, SAM2 [8], extended this capability to video with a streaming memory architecture that maintains temporal context through memory banks and cross-attention mechanisms. However, both models are designed for single-modal (RGB) input and focus on instance-level segmentation, requiring adaptation for multi-modal semantic tasks.

Low-Rank Adaptation (LoRA) [3] provides a parameter-efficient strategy for adapting large pretrained models by injecting trainable low-rank matrices into frozen attention layers. Applied to SAM, LoRA enables task-specific adaptation while preserving the foundation model’s generalization capability with minimal additional parameters.

MoE-LoRA [16] advanced this approach by incorporating Mixture-of-Experts (MoE) [10] routing into the LoRA framework for multi-modal SAM adaptation. By maintaining multiple expert pairs and using a gating network for top- k selection, it achieves modality-adaptive feature extraction with a 32.15% improvement under missing modality conditions. However, the hard top- k routing can lead to underutilized experts when the number of experts is small, which is common when the expert count matches the number of input modalities. Recent work on Soft Mixtures of Experts [7] has shown that replacing discrete top- k selection with continuous softmax weighting over all experts improves gradient flow and eliminates dead expert problems, motivating our adoption of soft gating in the LoRA context.

MemorySAM [1] proposed an alternative approach by repurposing SAM2’s video memory mechanism for multi-modal fusion. By treating different modalities as sequential frames, the memory bank naturally captures cross-modal features without requiring explicit fusion modules. Combined with a Semantic Prototype Memory Module (SPMM) that maintains category-level prototypes through exponential moving average updates, it bridges the gap from instance-level to semantic-level understanding, achieving state-of-the-art results of 65.38% mIoU on DELIVER and 52.88% on MCubeS.

Our work builds upon MemorySAM’s memory-based multi-modal framework while introducing Soft-MoE LoRA for improved expert utilization and a hybrid modulation strategy that separates absolute quality assessment from relative importance ranking.

3. Method

Our method adapts MemorySAM [1] for the MULTIAQUA maritime segmentation task. We retain the core idea of treating each modality as a sequential frame processed through SAM2’s [8] memory-based tracking pipeline, and modify the LoRA adaptation strategy, modality scoring, and output fusion. Our pipeline operates in four phases: given M input modalities (RGB, thermal, LiDAR), (1) all modalities are encoded by SAM2’s frozen image encoder with Soft-MoE LoRA layers, where an MI routing loss drives modality-specific expert specialization; (2) a cross-modal fusion head jointly compares the backbone features and produces softmax-normalized relative weights; (3) these weights are re-normalized with a higher temperature and scaled by M to modulate vision features before SAM2’s memory-augmented tracking step (UAMM); and (4) the per-modality mask outputs are fused using the raw softmax weights (AMF).

3.1. Soft-MoE LoRA with MI Specialization

We keep SAM2’s image encoder frozen and inject low-rank adaptation matrices into the query and value projections of each attention block, following the standard LoRA [3] formulation. Rather than using a single pair of low-rank matrices, we maintain N expert pairs $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^N$, where $\mathbf{A}_i \in \mathbb{R}^{d \times r}$ and $\mathbf{B}_i \in \mathbb{R}^{r \times d}$, with r being the LoRA rank and d the feature dimension.

Following the Soft Mixture-of-Experts formulation [7], instead of selecting the top- k experts via hard routing [10, 16], we compute a softmax-weighted combination over all experts for each input token $\mathbf{x}$:

$$\Delta \mathbf{x} = \sum_{i=1}^N \alpha_i(\mathbf{x}) \cdot \mathbf{B}_i \mathbf{A}_i \mathbf{x}, \quad \alpha_i(\mathbf{x}) = \frac{\exp(g_i(\mathbf{x}))}{\sum_{j=1}^N \exp(g_j(\mathbf{x}))}, \quad (1)$$

where $g(\mathbf{x}) = \mathbf{W}_g \mathbf{x} + \mathbf{b}_g$ is a lightweight gating network with $\mathbf{W}_g \in \mathbb{R}^{N \times d}$. The output of the original QKV projection is augmented as $\mathbf{q} \leftarrow \mathbf{q} + \Delta \mathbf{q}$ and $\mathbf{v} \leftarrow \mathbf{v} + \Delta \mathbf{v}$. When the number of experts N is small (e.g., $N=3$), hard top- k routing with $k=2$ permanently disables one expert per token; soft gating avoids this by ensuring gradient flow to every expert at every step. We use $N=3$ experts and $r=4$ throughout our experiments.

MI routing loss. However, soft gating introduces a different problem: without explicit specialization pressure, all experts receive similar gradients and the gate converges to a near-uniform distribution (e.g., $\alpha \approx [0.33, 0.33, 0.34]$ for all modalities). When every modality passes through the same weighted combination of experts, the resulting backbone features become modality-agnostic, making it impossible for the downstream fusion head to distinguish modality quality.

To address this, we maximize the mutual information (MI) between the modality identity and the expert selection. During encoding, we collect each modality's gate distribution $\alpha_i \in \mathbb{R}^N$ by averaging the softmax gate weights across all spatial positions and all Soft-MoE LoRA layers. The MI is decomposed as $I = H(\bar{\alpha}) - H(\alpha \mid \text{modality})$, where:

$$H(\alpha \mid \text{mod.}) = \frac{1}{M} \sum_{i=1}^M \left[- \sum_{n=1}^N \alpha_{i,n} \log \alpha_{i,n} \right], \quad (2)$$

$$H(\bar{\alpha}) = - \sum_{n=1}^N \bar{\alpha}_n \log \bar{\alpha}_n, \quad \bar{\alpha} = \frac{1}{M} \sum_{i=1}^M \alpha_i. \quad (3)$$

The MI routing loss minimizes the negative MI:

$$\mathcal{L}_{\text{MI}} = H(\alpha \mid \text{mod.}) - H(\bar{\alpha}). \quad (4)$$

Minimizing $\mathcal{L}_{\text{MI}}$ simultaneously reduces the conditional entropy (each modality concentrates on fewer experts) and increases the marginal entropy (all experts are utilized overall), driving modality-specific expert specialization while preventing expert collapse.

3.2. Adaptive Cross-Modal Fusion

Quality-aware scoring. A central challenge in multi-modal fusion is estimating the relative quality of each modality. Prior approaches [1] use independent per-modality heads (e.g., a CNN followed by sigmoid activation), but when backbone features are rich, the sigmoid saturates—all scores converge to ~ 1.0 regardless of actual quality differences. Moreover, global average pooling alone encodes semantic content rather than quality: a degraded RGB frame and a clear one may yield similar GAP vectors after backbone processing.

We address both issues by enriching each modality's representation with quality-sensitive statistics. Given backbone feature maps $\{\mathbf{F}_i\}_{i=1}^M$, we compute a multi-pool descriptor for each modality:

$$\mathbf{q}_i = [\text{GAP}(\mathbf{F}_i); \text{GMP}(\mathbf{F}_i); \sigma(\mathbf{F}_i)], \quad (5)$$

where GAP and GMP denote global average and max pooling respectively, and $\sigma(\mathbf{F}_i) \in \mathbb{R}^C$ is the channel-wise standard deviation. Each descriptor $\mathbf{q}_i \in \mathbb{R}^{3C}$ is compressed by a modality-specific network ϕ_i :

$$\mathbf{z}_i = \phi_i(\mathbf{q}_i) = \text{ReLU}(\mathbf{W}_c^{(i)} \cdot \mathbf{q}_i + \mathbf{b}_c^{(i)}), \quad (6)$$

where $\mathbf{W}_c^{(i)} \in \mathbb{R}^{h \times 3C}$ are modality-specific weights. The compressed vectors are concatenated and passed through a comparison MLP ψ to produce modality weights:

$$\mathbf{l} = \psi([\mathbf{z}_1; \dots; \mathbf{z}_M]), \quad w_i = \frac{\exp(l_i/\tau)}{\sum_{j=1}^M \exp(l_j/\tau)}, \quad (7)$$

where τ is a temperature parameter. The output layer of ψ is zero-initialized so that all modalities receive uniform weight $1/M$ at the start of training.

Gating oracle supervision. The segmentation objective alone provides an insufficient gradient signal to drive adaptive gating: a network assigning near-constant weights already minimizes the main loss because the downstream mask fusion can compensate. To provide a direct per-image training signal, we attach a lightweight auxiliary segmentation head $\mathcal{H}_i$ to each modality's backbone features. We derive oracle gating weights from their predictions:

$$\tilde{w}_i = \frac{\exp(-\mathcal{L}_{\text{aux}}^{(i)})}{\sum_{j=1}^M \exp(-\mathcal{L}_{\text{aux}}^{(j)})}, \quad (8)$$

where $\mathcal{L}_{\text{aux}}^{(i)}$ is the per-image cross-entropy loss for modality i 's auxiliary head. The fusion head is trained to match these oracle weights via KL divergence:

$$\mathcal{L}_{\text{gate}} = D_{\text{KL}}(\tilde{\mathbf{w}} \parallel \mathbf{w}), \quad (9)$$

where $\tilde{\mathbf{w}}$ is detached from the computation graph. Combined with the MI routing loss that differentiates backbone features across modalities (Sec. 3.1), the oracle supervision ensures that the fusion head learns to translate these feature differences into input-adaptive gating weights.

UAMM and AMF. The cross-modal weights are applied in two complementary paths. For *Uncertainty-Aware Memory Modulation* (UAMM), the raw logits $\mathbf{l}$ are re-normalized with a higher temperature τ_u and scaled by M :

$$\hat{w}_i = M \cdot \frac{\exp(l_i/\tau_u)}{\sum_{j=1}^M \exp(l_j/\tau_u)}, \quad \hat{\mathbf{f}}_i^{(k)} = \hat{w}_i \cdot \mathbf{f}_i^{(k)}, \quad (10)$$

where $\tau_u=2.0$. The M -scaling preserves the average feature magnitude at 1.0: when all modalities are equally reliable, $\hat{w}_i = 1.0$ for all; when one degrades, its features are suppressed while the best modality is amplified. The modulated features are processed by SAM2's `track_step`, where the first modality initializes the memory and subsequent modalities attend to it via cross-attention.

For *Adaptive Mask Fusion* (AMF), the per-modality mask predictions $\mathbf{o}_i$ from the tracking loop are fused using the raw softmax weights:

$$\mathbf{o}_{\text{fused}} = \sum_{i=1}^M w_i \cdot \mathbf{o}_i. \quad (11)$$

UAMM uses a softer distribution ($\tau_u > \tau$) to gently redistribute feature magnitudes, while AMF uses the sharper distribution from τ for decisive output weighting.

3.3. Training Objectives

Following MemorySAM [1], we use a Semantic Prototype Memory Module: for each class c , a global prototype $\mathbf{p}_c$

is maintained via exponential moving average (momentum = 0.8), and the prototype loss is $\mathcal{L}_{\text{proto}} = \text{MSE}(\hat{\mathbf{p}}, \mathbf{p})$. The total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{OHEM}} + \mathcal{L}_{\text{proto}} + \lambda_{\text{gate}} \left(\mathcal{L}_{\text{gate}} + \gamma \cdot \frac{1}{M} \sum_i \mathcal{L}_{\text{aux}}^{(i)} \right) + \lambda_{\text{MI}} \cdot \mathcal{L}_{\text{MI}} \quad (12)$$

where $\mathcal{L}_{\text{OHEM}}$ is the Online Hard Example Mining cross-entropy loss, $\gamma=0.3$, $\lambda_{\text{gate}}=0.5$, and $\lambda_{\text{MI}}=1.0$.

3.4. Night-Robust RGB Augmentation

The MULTIAQUA dataset requires generalization from daytime training to nighttime evaluation, where RGB quality degrades severely while thermal and LiDAR remain relatively stable. Inspired by [6] and CRM [11], we apply three complementary RGB corruptions during training (leaving thermal and LiDAR unchanged):

Night Simulation (probability 0.35): extreme brightness reduction (factor sampled from $[0.03, 0.25]$), followed by contrast reduction ($[0.3, 0.7]$), gamma correction ($\gamma \in [0.4, 0.8]$), and additive Gaussian noise ($\sigma = 0.02$).

Complementary Random Masking (probability 0.30): rectangular patches are randomly placed on the RGB image, zeroing out 20–50% of spatial regions, forcing the model to rely on thermal and LiDAR for the masked areas.

Full-Channel Zeroing (probability 0.12): the entire RGB input is replaced with zeros, simulating complete RGB failure.

These augmentations are stacked sequentially: an image may undergo night simulation and then additionally be partially masked, creating diverse degradation patterns.

4. Experiments

4.1. Dataset and Setup

Dataset. We evaluate on the MULTIAQUA dataset [6], which provides synchronized RGB, thermal, and LiDAR data captured from unmanned surface vehicles. The dataset contains four semantic classes: static obstacle, dynamic obstacle, water, and sky. Models are trained on daytime data and evaluated on both a daytime validation set and a nighttime test set, where RGB quality degrades significantly.

Implementation details. We use SAM2 (Hiera-based backbone) as the foundation model with LoRA rank $r=4$ and $N=3$ Soft-MoE experts. All images are resized to 1024×1024 . We train for 200 epochs with AdamW optimizer (learning rate 6×10^{-4} , weight decay 0.01) using a warmup-polynomial learning rate schedule (10 warmup epochs, power 0.9). The batch size is 1 per GPU with gradient accumulation, and we use Online Hard Example Mining (OHEM) cross-entropy loss with the prototype segmentation loss (??). SAM2’s image encoder is frozen; only the Soft-MoE LoRA layers, cross-modal fusion head, prompt

Table 1. Main results on the MULTIAQUA benchmark. Daytime val mIoU and nighttime test mIoU are reported. [†]Results from the MaCVi challenge leaderboard.

Method	Modalities	Val mIoU	Test mIoU
CMNeXt [15]	RGB+T+L	[TODO: XX.X]	[TODO: XX.X]
Ours	RGB+T+L	94.00	[TODO: XX.X]

Table 2. Per-class IoU (%) on the MULTIAQUA validation set.

	Static	Dynamic	Water	Sky	mIoU
Ours	97.25	80.82	99.08	98.84	94.00

Table 3. Ablation on night-robust RGB augmentation. “Night Aug” includes night simulation, CRM, and full-channel zeroing.

Night Aug	Val mIoU	Test mIoU
✗	[TODO: XX.X]	[TODO: XX.X]
✓	94.00	[TODO: XX.X]

encoder, and mask decoder are trained. Training is conducted with PyTorch DDP on NVIDIA GPUs.

Evaluation metric. We report mean Intersection-over-Union (mIoU) averaged over the four semantic classes.

4.2. Main Results

Tab. 1 presents our results on the MULTIAQUA benchmark. On the daytime validation set, our method achieves 94.00% mIoU.

Tab. 2 shows the per-class breakdown on the validation set. The model achieves strong performance on water (99.08%) and sky (98.84%), which occupy large contiguous regions, and on static obstacles (97.25%). The dynamic obstacle class is the most challenging at 80.82%, which is expected given the smaller object sizes and higher appearance variability of moving vessels.

4.3. Ablation Studies

Effect of night augmentation. To validate the importance of our night-robust augmentation pipeline (Sec. 3.4), we train an identical model without any RGB corruption augmentations and compare performance.

Effect of modality combination. We investigate the contribution of each modality by training models with different subsets.

5. Conclusion

We presented a method for adapting foundation models to multi-modal maritime semantic segmentation under varying sensor conditions. By building on MemorySAM’s

Table 4. Ablation on input modalities.

Modalities	Val mIoU	Test mIoU
RGB only	[TODO: XX.X]	[TODO: XX.X]
RGB + Thermal	[TODO: XX.X]	[TODO: XX.X]
RGB + Thermal + LiDAR	94.00	[TODO: XX.X]

memory-based cross-modal framework, we identified and addressed two key failure modes: (1) soft MoE gating converges to uniform expert routing, producing modality-agnostic features—resolved by an MI routing loss that drives modality-specific expert specialization; and (2) the segmentation objective alone cannot train adaptive fusion weights—resolved by quality-aware cross-modal scoring with gating oracle supervision via per-modality auxiliary heads. Combined with aggressive RGB corruption augmentations for day-to-night generalization, our experiments on the MULTIAQUA benchmark validate each design choice and demonstrate the importance of explicit training supervision for both expert routing and modality gating in degraded sensing conditions.

References

- [1] Yuqi Chen et al. MemorySAM: Memorize modalities and semantics with SAM2 for multi-modal semantic segmentation. *arXiv preprint arXiv:2503.06700*, 2025. 1, 2, 3, 4
- [2] Caner Hazirbas, Lingni Ma, Csaba Domokos, and Daniel Cremers. FuseNet: Incorporating depth as input and output. In *BMVC*, 2017. 2
- [3] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022. 1, 3
- [4] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *ICCV*, pages 4015–4026, 2023. 1, 3
- [5] Bingyu Li, Da Zhang, Zhiyuan Zhao, Junyu Gao, and Xue-long Li. StitchFusion: Weaving any visual modalities to enhance multimodal semantic segmentation. *arXiv preprint arXiv:2408.01343*, 2024. 1, 2
- [6] Jon Muhovič and Janez Perš. MULTIAQUA: A multimodal maritime dataset and robust training strategies for multimodal semantic segmentation. *arXiv preprint arXiv:2512.17450*, 2025. 1, 2, 5
- [7] Joan Puigcerver, Carlos Riquelme, Basil Mustafa, and Neil Houlsby. From sparse to soft mixtures of experts. 2024. 2, 3
- [8] Nikhila Ravi, Valentin Gabber, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. 1, 3
- [9] Md Kaykobad Reza, Ashley Prater-Bennette, and M Salman Asif. MMSFormer: Multimodal transformer for material and semantic segmentation. *IEEE Open Journal of Signal Processing*, 2024. 1, 2
- [10] Noam Shazeer, Azalia Mirhoseini, Krzysztof Mazyar, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *ICLR*, 2017. 3
- [11] Ukcheol Shin, Kyunghyun Lee, In So Kweon, and Jean Oh. Complementary random masking for RGB-thermal semantic segmentation. In *IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024. 1, 2, 5
- [12] Yuxiang Sun, Weixun Zuo, and Ming Liu. RTFNet: Rgb-thermal fusion network for semantic segmentation of urban scenes. In *IEEE Robotics and Automation Letters*, pages 2576–2583, 2019. 2
- [13] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. SegFormer: Simple and efficient design for semantic segmentation with transformers. In *NeurIPS*, 2021. 2
- [14] Jiaming Zhang, Huayao Liu, Kailun Yang, Xinxin Hu, Ruiping Liu, and Rainer Stiefelhagen. CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers. *IEEE Trans. Intell. Transp. Syst.*, 24(12):14679–14694, 2023. 1, 2
- [15] Jiaming Zhang, Ruiping Liu, Hao Shi, Kailun Yang, Simon Reiß, Kunyu Peng, Haodong Fu, Kaiwei Wang, and Rainer Stiefelhagen. Delivering arbitrary-modal semantic segmentation. In *CVPR*, pages 1136–1147, 2023. 1, 2, 5
- [16] Chenyang Zhu, Bin Xiao, Lin Shi, Shoukun Xu, and Xu Zheng. Customize segment anything model for multi-modal semantic segmentation with mixture of LoRA experts. *arXiv preprint arXiv:2412.04220*, 2024. 1, 2, 3